Z-Axis Actuator Sizing

The Z axis of the Cartesian robotic arm is driven by a stepper motor and a power-screw (lead-screw) mechanism. It raises and lowers the suction cup so that it can reach and pick up the product parts easily.

The power screw is a **Tr8x8** Acme (trapezoidal) screw, **173 mm** long. In this designation the first 8 is the 8 mm outer diameter and the second 8 is the lead, **L = 8 mm**, since the thread has a **pitch p = 2 mm** with **4 starts** (L = n × p = 4 × 2 mm). The lead L is used in the inertia, direct torque and lead-angle equations, while the pitch p is used only in the root diameter equation.

Governing Formulas

1. Torque Equations

- Total Motor Torque:

$$T_m = T_r + J_{\text{eq}} \cdot \ddot{\theta}_m$$

- Peak Required Motor Torque:

$$T_{m,\text{peak}} = \max\left(T_{m,\text{acc}}, T_{m,\text{const}}, T_{m,\text{dec}}\right)$$

- RMS Torque (T_{rms}):

$$T_{\text{rms}} = \sqrt{\frac{T_{m,\text{acc}}^2 \cdot t_{\text{acc}} + T_{m,\text{const}}^2 \cdot t_{\text{const}} + T_{m,\text{dec}}^2 \cdot t_{\text{dec}} + T_{m,\text{dw}}^2 \cdot t_{\text{dw}}}{t_{\text{acc}} + t_{\text{const}} + t_{\text{dec}} + t_{\text{dw}}}}$$

2. Inertia Equations

- Load Rotary Inertia (J_{load}):

$$J_{\text{load}} = \frac{m_{\text{load}}}{\eta \cdot \left(\frac{2\pi}{L}\right)^2} + \frac{1}{2} \cdot m_{sc} \cdot r^2$$

- Motor Rotary Inertia (J_{motor}):

$$J_{\text{motor}} = \frac{1}{10} \cdot J_{\text{load}}$$

- Equivalent System Inertia (J_{eq}):

$$J_{\text{eq}} = J_{\text{load}} + J_{\text{motor}}$$

3. Resisting Torque Equations

- Total Resisting Torque (T_r):

$$T_r = \frac{F_{\text{load}}}{\eta \cdot \left(\frac{2\pi}{L}\right)} + T_{\text{friction}}$$

- Thread Friction Torque (T_{friction}):

$$T_{\text{friction}} = \left(\frac{F \cdot d_m}{2}\right) \cdot \left(\frac{\tan(\alpha) + \mu'}{1 - \mu' \cdot \tan(\alpha)}\right)$$

- Friction Coefficient (μ'):

$$\mu' = \frac{\mu}{\cos(\beta)}$$

- Screw Mean Diameter (d_m), Root Diameter (d_{root}) & Lead Angle (α):

$$d_m = \frac{d_{\text{outer}} + d_{\text{root}}}{2} + 0.25$$

$$d_{\text{root}} = d_{\text{outer}} - p - 0.5$$

$$\alpha = \arctan\left(\frac{L}{\pi \cdot d_m}\right)$$

4. Acceleration Equations

- Maximum Angular Velocity (ω_{max}):

$$v_{\text{max}} = \omega_{\text{max}} \cdot \left(\frac{L}{2\pi}\right)$$

- Total Rotation, Area Under the Curve (θ):

$$\theta = \frac{1}{2} t_{\text{acc}} \omega_{\text{max}} + \omega_{\text{max}} t_{\text{const}} + \frac{1}{2} t_{\text{dec}} \omega_{\text{max}}$$

- Acceleration Phase ($\ddot{\theta}_{\text{acc}}$):

$$\ddot{\theta}_{\text{acc}} = \frac{\omega_{\text{max}} - 0}{t_{\text{acc}}}$$

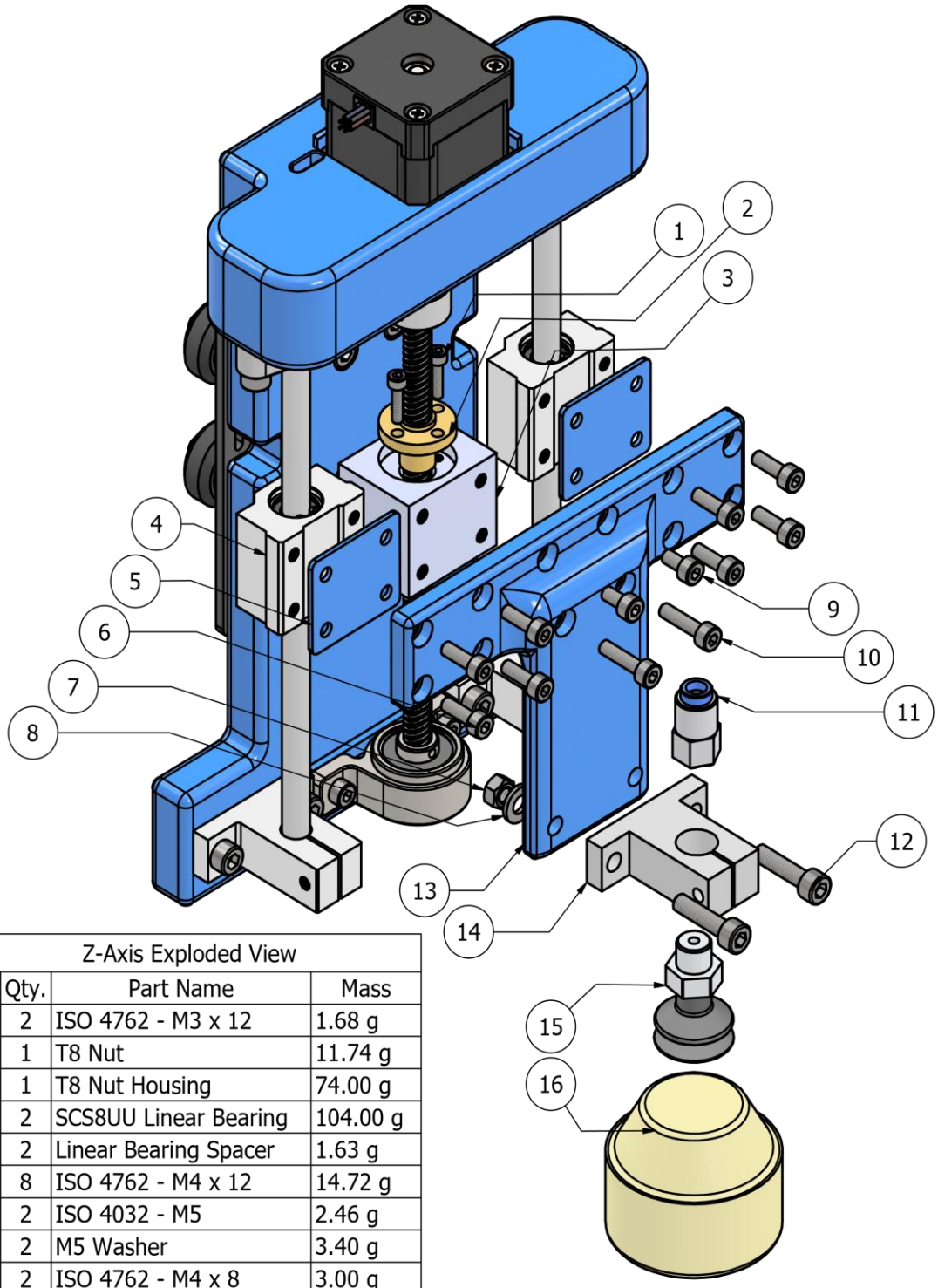
- Constant-Speed Phase ($\ddot{\theta}_{\text{const}}$):

$$\ddot{\theta}_{\text{const}} = 0$$

- Deceleration Phase ($\ddot{\theta}_{\text{dec}}$):

$$\ddot{\theta}_{\text{dec}} = \frac{0 - \omega_{\text{max}}}{t_{\text{dec}}}$$

To size the actuator correctly, the total mass carried by the power screw is determined first. Summing all moving parts in the Z-axis bill of materials gives a total mass of **369.58 g**.



Z-Axis Exploded View			
No.	Qty.	Part Name	Mass
1	2	ISO 4762 - M3 x 12	1.68 g
2	1	T8 Nut	11.74 g
3	1	T8 Nut Housing	74.00 g
4	2	SCS8UU Linear Bearing	104.00 g
5	2	Linear Bearing Spacer	1.63 g
6	8	ISO 4762 - M4 x 12	14.72 g
7	2	ISO 4032 - M5	2.46 g
8	2	M5 Washer	3.40 g
9	2	ISO 4762 - M4 x 8	3.00 g
10	4	ISO 4762 - M4 x 16	4.50 g
10	4	ISO 4762 - M5 x 16	6.90 g
11	1	Pneumatic Fitting	7.00 g
12	2	ISO 4762 - M5 x 20	7.20 g
13	1	Suction Cup Fixation	56.83 g
14	1	SK12 Shaft Holder	30.00 g
15	1	Suction Cup	8.00 g
16	1	Product Base	32.52 g

1. Inertia Calculation (J_{eq})

Variables & Parameters:

- Load Mass: $m_{load} = 369.58\ \text{g} = 0.36958\ \text{kg}$
- Mechanical Efficiency (assumed): $\eta = 65\% = 0.65$
- Lead: $L = 8\ \text{mm} = 0.008\ \text{m}$
- Screw: Tr8x8, $d = 8\ \text{mm} \Longrightarrow r = 0.004\ \text{m}$, $\ell = 0.173\ \text{m}$, $\rho = 7850\ \text{kg/m}^3$ (steel)

Equations & Substitution:

- Load Rotary Inertia (J_{load}):

$$J_{load} = \frac{m_{load}}{\eta \cdot \left(\frac{2\pi}{L}\right)^2}$$

$$J_{load} = \frac{0.36958\ \text{kg}}{0.65 \cdot \left(\frac{2\pi}{0.008\ \text{m}}\right)^2} = \frac{0.36958}{0.65 \cdot (785.3982)^2} = \frac{0.36958}{400{,}952.68}$$

$$J_{load} = 9.21755 \times 10^{-7}\ \text{kg} \cdot \text{m}^2$$

- Lead Screw Inertia (J_{screw}):

$$m_{sc} = \rho \cdot \pi \cdot r^2 \cdot \ell = 7850 \cdot \pi \cdot (0.004)^2 \cdot 0.173 = 0.06826\ kg$$

$$J_{screw} = \frac{1}{2} \cdot m_{sc} \cdot r^2$$

$$J_{screw} = \frac{1}{2} \cdot 0.06826\ kg \cdot (0.004\ m)^2 = 5.461 \times 10^{-7}\ kg \cdot m^2$$

- Motor Inertia (J_{motor}):

$$J_{motor} = \frac{1}{10} \cdot (J_{load} + J_{screw})$$

$$J_{motor} = \frac{1}{10} \cdot (9.21755 \times 10^{-7}\ \text{kg} \cdot \text{m}^2 + 5.461 \times 10^{-7}\ \text{kg} \cdot \text{m}^2) = 1.46786 \times 10^{-7}\ \text{kg} \cdot \text{m}^2$$

- Equivalent System Inertia (J_{eq}):

$$J_{eq} = J_{load} + J_{screw} + J_{motor}$$

$$J_{eq} = 9.21755 \times 10^{-7}\ \text{kg} \cdot \text{m}^2 + 5.461 \times 10^{-7}\ \text{kg} \cdot \text{m}^2 + 1.46786 \times 10^{-7}\ \text{kg} \cdot \text{m}^2$$

$$J_{eq} = 1.61464 \times 10^{-6}\ \text{kg} \cdot \text{m}^2$$

2. Resisting Torque Calculation (T_r)

Variables & Parameters:

- Gravitational Acceleration: $g = 9.81\ \text{m/s}^2$
- Vertical Load Force: $F_{\text{load}} = W = m_{\text{load}} \cdot g = 0.36958\ \text{kg} \times 9.81\ \text{m/s}^2 = 3.62558\ \text{N}$
- Outer Diameter: $d_{\text{outer}} = 8\ \text{mm} = 0.008\ \text{m}$
- Pitch: $p = 2\ \text{mm} = 0.002\ \text{m}$
- Lead: $L = 8\ \text{mm} = 0.008\ \text{m}$
- Thread Angle: $\beta = 15°$
- Friction Coefficient (Brass Nut on Steel Screw): $\mu = 0.2$
- Mechanical Efficiency (assumed): $\eta = 0.65$

Intermediate Geometry & Friction Calculations:

- Modified Friction Coefficient (μ'):

$$\mu' = \frac{\mu}{\cos(\beta)} = \frac{0.2}{\cos(15°)} = \frac{0.2}{0.96593} = 0.20706$$

- Root Diameter (d_{root}):

$$d_{\text{root}} = d_{\text{outer}} - p - 0.5 = 8\ \text{mm} - 2\ \text{mm} - 0.5\ \text{mm} = 5.50\ \text{mm} = 0.0055\ \text{m}$$

- Mean Diameter (d_m / d_{mean}):

$$d_m = \frac{d_{\text{outer}} + d_{\text{root}}}{2} + 0.25 = \frac{8 + 5.50}{2} + 0.25 = 6.75 + 0.25 = 7.00\ \text{mm} = 0.0070\ \text{m}$$

- Lead Angle (α):

$$\alpha = \arctan\left(\frac{L}{\pi \cdot d_m}\right) = \arctan\left(\frac{0.008\ \text{m}}{\pi \cdot 0.007\ \text{m}}\right) = \arctan(0.36378) = 19.9905°$$

$$\tan(\alpha) = 0.36378$$

Torque Equations & Substitution:

- Thread Friction Torque (T_{friction}):

$$T_{\text{friction}} = \left(\frac{F \cdot d_m}{2}\right) \cdot \left(\frac{\tan(\alpha) + \mu'}{1 - \mu' \cdot \tan(\alpha)}\right)$$

$$T_{\text{friction}} = \left(\frac{3.62558\ \text{N} \cdot 0.0070\ \text{m}}{2}\right) \cdot \left(\frac{0.36378 + 0.20706}{1 - 0.20706 \cdot 0.36378}\right)$$

$$T_{\text{friction}} = 0.0078337\ \text{N} \cdot \text{m}$$

- Direct Load Lifting Torque (T_{direct}):

$$T_{\text{direct}} = \frac{F_{\text{load}}}{\eta \cdot \left(\frac{2\pi}{L}\right)} = \frac{F_{\text{load}} \cdot L}{2\pi \cdot \eta}$$

$$T_{\text{direct}} = \frac{3.62558\ \text{N} \cdot 0.008\ \text{m}}{2\pi \cdot 0.65} = 0.0071019\ \text{N} \cdot \text{m}$$

- Total Resisting Torque (T_r):

$$T_r = T_{\text{direct}} + T_{\text{friction}}$$

$$T_{\text{r}} = 0.0071019\ \text{N} \cdot \text{m} + 0.0078337\ \text{N} \cdot \text{m} = 0.0149356\ \text{N} \cdot \text{m}$$

3.Acceleration ($\ddot{\theta_m}$):

The stroke of the Z axis is **95 mm** and the desired maximum nut speed is **65 mm/s**, so at constant speed the whole stroke would take about **1.46 s** (95 ÷ 65). The acceleration and deceleration ramps of the velocity profile make the actual move a little longer, as calculated below.

Variables & Parameters:

- Stroke (Worst Case): $S = 95\ \text{mm} = 0.095\ \text{m}$
- Lead: $L = n \times p = 4 \times 2\ \text{mm} = 8\ \text{mm} = 0.008\ \text{m}$
- Maximum Linear Speed (desired): $v_{\text{max}} = 65\ \text{mm/s} = 0.065\ \text{m/s}$
- Phase Time Ratio (assumed, symmetric profile): $t_{\text{acc}} : t_{\text{const}} : t_{\text{dec}} = 0.3 : 0.4 : 0.3$ of the move time T
- Dwell Time (assumed): $t_{\text{dw}} = 0.2\ \text{s}$
- No gearbox $\Longrightarrow \theta_m = \theta_L$

Equations & Substitution:

- Maximum Angular Velocity (ω_{max}):

$$\omega_{\text{max}} = \frac{v_{\text{max}}}{\left(\frac{L}{2\pi}\right)}$$

$$\omega_{\text{max}} = \frac{0.065\ \text{m/s}}{\left(\frac{0.008\ \text{m}}{2\pi}\right)} = \frac{0.065}{0.0012732} = 51.0509\ \text{rad/s} \approx 487.5\ \text{rpm}$$

- Total Rotation for the 95 mm Stroke (θ):

$$\theta = \frac{S}{\left(\frac{L}{2\pi}\right)} = \frac{0.095\ \text{m}}{\left(\frac{0.008\ \text{m}}{2\pi}\right)} = \frac{0.095}{0.0012732} = 74.6128\ \text{rad} \approx 11.875\ \text{rev}$$

- Phase Times, $t_{acc}: t_{const}: t_{dec} = 0.3: 0.4: 0.3$ of the total move time T:

$$\theta = \frac{1}{2}(0.3T)\omega_{max} + \omega_{max}(0.4T) + \frac{1}{2}(0.3T)\omega_{max} = 0.7 \cdot \omega_{max} \cdot T$$

$$T = \frac{\theta}{0.7 \cdot \omega_{max}} = \frac{74.6128}{0.7 \cdot 51.0509} = \frac{74.6128}{35.7356} = 2.0879 \text{ s}$$

$$t_{acc} = 0.3 \cdot T = 0.3 \cdot 2.0879 = 0.6264 \text{ s}$$

$$t_{const} = 0.4 \cdot T = 0.4 \cdot 2.0879 = 0.8352 \text{ s}$$

$$t_{dec} = 0.3 \cdot T = 0.3 \cdot 2.0879 = 0.6264 \text{ s}$$

$$\text{Cycle time } = T + t_{dw} = 2.0879 + 0.2 = 2.2879 \text{ s}$$

- Acceleration Phase ($\ddot{\theta}_{acc}$):

$$\ddot{\theta}_{acc} = \frac{\omega_{max} - 0}{t_{acc}} = \frac{51.0509 - 0}{0.6264} = 81.50 \text{ rad/s}^2$$

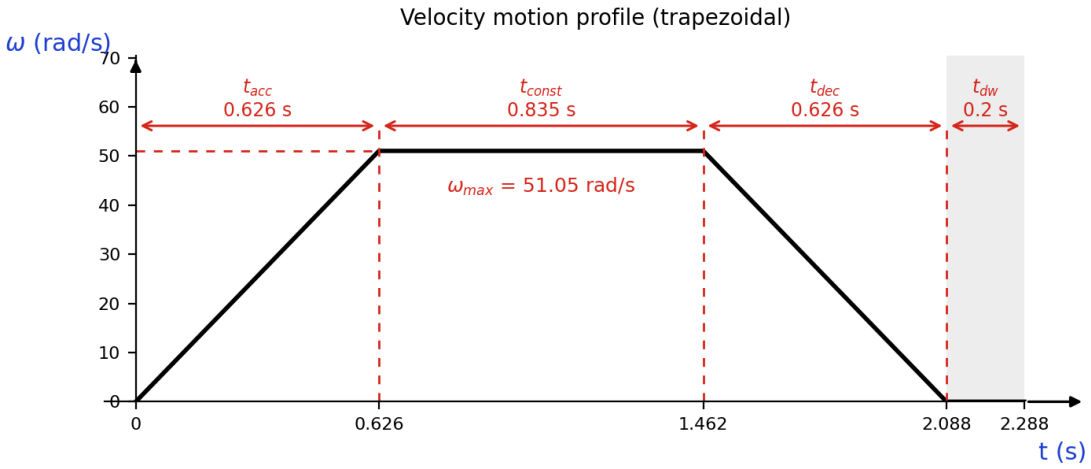
- Constant-Speed Phase ($\ddot{\theta}_{const}$):

$$\ddot{\theta}_{const} = 0 \text{ rad/s}^2$$

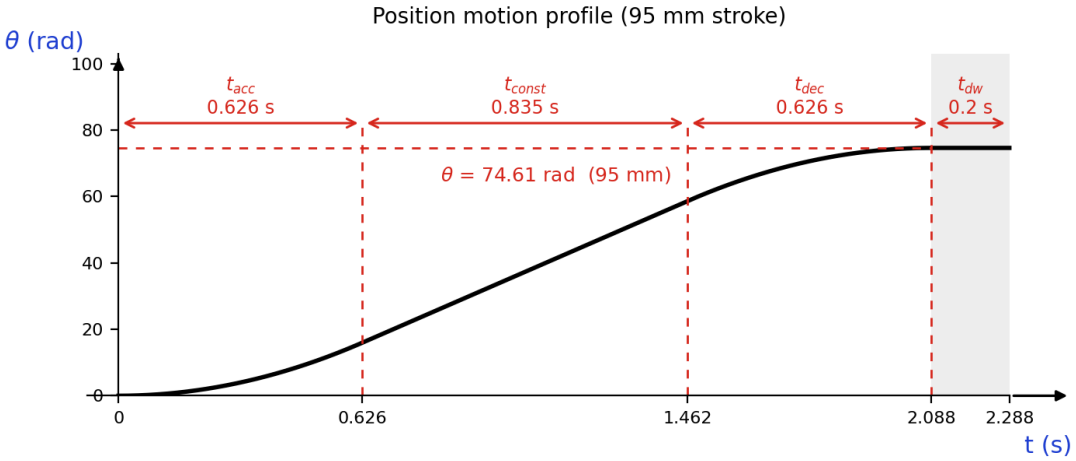
- Deceleration Phase ($\ddot{\theta}_{dec}$):

$$\ddot{\theta}_{dec} = \frac{0 - \omega_{max}}{t_{dec}} = \frac{0 - 51.0509}{0.6264} = -81.50 \text{ rad/s}^2$$

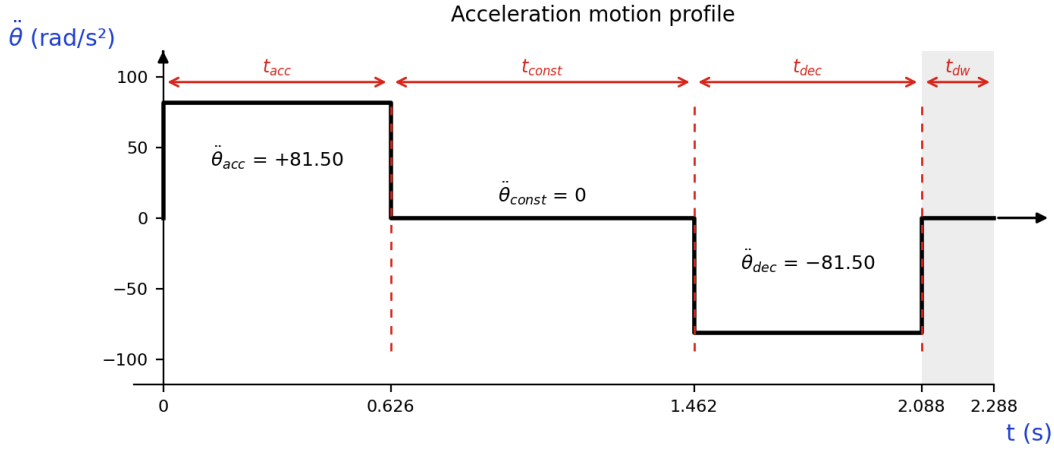
- Velocity Motion Profile (ω vs t):



- Position Motion Profile (θ vs t):



- Acceleration Motion Profile ($\ddot{\theta}$ vs t):



4. Motor Torque Calculations (T_m)

Variables & Parameters:

- Equivalent System Inertia: $J_{eq} = 1.61464 \times 10^{-6}$ kg · m^2
- Total Resisting Torque: $T_r = 0.0149356$ N · m
- Acceleration Phase Angular Acceleration: $\ddot{\theta}_{acc} = +81.50$ rad/s^2
- Constant-Speed Phase Angular Acceleration: $\ddot{\theta}_{const} = 0$ rad/s^2
- Deceleration Phase Angular Acceleration: $\ddot{\theta}_{dec} = -81.50$ rad/s^2

Torque Calculations & Substitution:

1. Acceleration Phase Motor Torque ($T_{m,acc}$):

$$T_{m,acc} = T_r + J_{eq} \cdot \ddot{\theta}_{acc}$$

$$T_{m,acc} = 0.0149356 \text{ N} \cdot \text{m} + \left(1.61464 \times 10^{-6} \text{ kg} \cdot \text{m}^2 \times 81.50 \text{ rad/s}^2\right)$$

$$T_{m,acc} = 0.0149356 + 0.00013159 = 0.0150672 \text{ N} \cdot \text{m}$$

2. Constant-Speed Phase Motor Torque ($T_{m,const}$):

$$T_{m,const} = T_r + J_{eq} \cdot \ddot{\theta}_{const}$$

$$T_{m,const} = 0.0149356 \text{ N} \cdot \text{m} + \left(1.61464 \times 10^{-6} \text{ kg} \cdot \text{m}^2 \times 0 \text{ rad/s}^2\right) = 0.0149356 \text{ N} \cdot \text{m}$$

3. Deceleration Phase Motor Torque ($T_{m,dec}$):

$$T_{m,dec} = T_r + J_{eq} \cdot \ddot{\theta}_{dec}$$

$$T_{m,dec} = 0.0149356 \text{ N} \cdot \text{m} + \left(1.61464 \times 10^{-6} \text{ kg} \cdot \text{m}^2 \times \left(-81.50 \text{ rad/s}^2\right)\right)$$

$$T_{m,dec} = 0.0149356 - 0.00013159 = 0.0148040 \text{ N} \cdot \text{m}$$

4. Dwell Phase Motor Torque ($T_{m,dw}$):

$$T_{m,dw} = T_{m,const} = 0.0149356 \text{ N} \cdot \text{m}$$

Peak and RMS Torque Selection:

- Peak Motor Torque ($T_{m,peak}$):

$$T_{m,peak} = \max\left(T_{m,acc}, T_{m,const}, T_{m,dec}\right)$$

$$T_{m,peak} = \max(0.0150672, 0.0149356, 0.0148040) = 0.0150672 \text{ N} \cdot \text{m}$$

- Root Mean Square Torque (T_{rms}):

$$T_{rms} = \sqrt{\frac{T_{m,acc}^2 \cdot t_{acc} + T_{m,const}^2 \cdot t_{const} + T_{m,dec}^2 \cdot t_{dec} + T_{m,dw}^2 \cdot t_{dw}}{t_{acc} + t_{const} + t_{dec} + t_{dw}}}$$

$$T_{rms} = \sqrt{\frac{(0.0150672)^2 \cdot 0.6264 + (0.0149356)^2 \cdot 0.8352 + (0.0148040)^2 \cdot 0.6264 + (0.0149356)^2 \cdot 0.20}{2.288}}$$

$$T_{rms} = \sqrt{\frac{0.00014220 + 0.00018631 + 0.00013728 + 0.00004461}{2.288}} = 0.0149358 \text{ N} \cdot \text{m}$$

5. Motor Selection & Verification with Safety Factor:

- Safety Factor Application ($n = 4$):

$$\text{Design RMS Torque } (T_{req,rms}) = n \cdot T_{m,rms} = 4 \times 0.0149358 \text{ N} \cdot \text{m}$$
$$= 0.0597432 \text{ N} \cdot \text{m} \quad (5.97 \text{ N} \cdot \text{cm})$$

- Selected Stepper Motor: NEMA 17 (17HS4401)



Model	NEMA 17 (17HS4401)
Step Angle	1.8°
Holding Torque	0.40 N·m (4.0 kg·cm)
Detent Torque	2.2 N·cm (0.022 N·m)
Rotor Inertia	54 g·cm²
Rated Current	1.3 A
Resistance	2.6 Ω
Inductance	4.5 mH
Leads	4 Leads
Motor Length	40 mm
Shaft Diameter	5 mm

Suitability Justification: The selected NEMA 17 17HS4401 motor can easily handle the maximum dynamic load requirement during motion (5.97 N · cm RMS torque requirement). Furthermore, its unpowered detent torque of 2.20 N · cm exceeds the continuous static gravity load (1.49 N · cm), ensuring that the lead-screw nut will not slip or back-drive when the robot is powered off.

In addition, the selected motor can reach the required maximum speed of 51 rad/s (≈486 RPM), which corresponds to about 1,620 full steps/s for a 200 steps/rev motor. At this speed the motor's pull-out torque stays above the 5.97 N·cm RMS requirement, with the driver supplied at 12 V.

Note: Even after updating the system inertia with the actual datasheet value of the motor (5.40×10^{-6} kg · m²), the recalculated peak torque with safety factor increases only slightly to 6.20 N · cm. The motor's rated holding torque of 40 N · cm is still nearly 6.5 × higher than this new requirement, confirming that the NEMA 17HS4401 remains capable of driving the required load with a safety margin.

Reference for the sizing equations: Industrial Monitor Direct, “Sizing Lead Screws and Ball Screws for Linear Actuation”. https://industrialmonitordirect.com/blogs/knowledgebase/sizing-lead-screws-and-ball-screws-for-linear-actuation

Reference for the speed equations: Firgelli Automations, “Screw Lead, RPM, and Travel Calculator”. https://www.firgelliauto.com/blogs/engineering-calculators/screw-lead-rpm-and-travel-calculator#related-resources